CS 534 Phase 4 - Project Final Paper and its Product (35 Points)
(Due: 11:59 p.m. on July 31st, 2025)

Deliverables: A zip file includes your final paper, all .py codes, and data used in your project.

Your project paper should be **AT LEAST 10 pages, not including the reference page,** to describe and explain the following sections, answers, and elements. You may include any figures, graphs, and tables to supplement your explanations and descriptions in your project paper if needed. Please feel free to ***reuse/update/modify*** all of the written material about your project that you have been preparing before for **Phase 1 – Project Proposal** and **Phase 2 Project Progress Report**. **However, please DO make sure that any omissions or problems that had been pointed out in the comments of Phase 1 and Phase 2 have now been fixed or points will be deducted accordingly.**

If your team works on **Option 1 - State-Of-The-Art AI Project**, please follow the instructions of Sections A1 below. If your team works on **Option 2 - Innovative/Advanced AI Project**, please read the instructions of Sections A2 below.

Note: (1) Your project should be related to AI topic areas; (2) Each group only submits one PDF copy; and (3) Each page format should be 1-inch Margin, Times New Roman, 10 pt, and Single Space.

(A1) Written Final Project Report Content

Option 1 - State-Of-The-Art AI Project

Project Title (1 Point)
Team Members (1 Point)
Photos (1 Point)
Current Programs (1 Point)

Abstract (200 words) (2 Points)

It summarizes the major aspects of the entire AI project in a prescribed sequence that includes:

- The overall purpose of the AI project that your group is working on is to solve a problem.
- The approach to address the problem in your project.
- The current major results and milestones of your project

Keywords (5-10 words or phrases) (1 Point)

Words or phrases in concepts are important in your project.

Section 1. Introduction (At Least 1 Page) (3 Points)

1 Paragraph

- What are the motivational background and context of your domain problem addressed by your project?
- Why is this domain problem important to solve? Please provide some real-world examples and significant statistics with citations to support the importance of this domain problem.

At LEAST 2 Paragraphs

- Describe and explain **at least two** (if your team has two members) or **at least three** (if your team has three members) current state-of-the-art (SOTA) methods with citations that should be able to address the domain problem stated in the above paragraph(s)?
- Please explain what **advantages** of those methods are and why those methods should be able to address the problem.

Note: Those existing SOTA methods should be the methods that have been applied to solve the problems described in the most recent papers, i.e., **from 2020 to present**.

At LEAST 2 Paragraphs

- What is your proposed high-level solution and process to solve the problem using the SOTA methods?
- How will you evaluate the effectiveness of those SOTA approach(es) listed above for your problem? That is, you need to describe how you will measure the performance or success of those current SOTA approaches described above that you can **replicate** and compare those SOTA methods in terms of the performance and select the best one among them?
- What data, experiments, and metrics will you use in your SOTA performance comparison?

1 Paragraph: Outline the subsequent sections that you will include for the rest of your report.

Section 2. SOTA Literature Review (At Least 0.5 Page) (4 Points)

Use **at least one or two paragraphs** to deeply describe each SOTA method that you mention in Section 1. For example, if there are three SOTA methods listed in Section 1, you should have at least three paragraphs, i.e., one paragraph with a workflow diagram for each method. Please also re-state what advantages of those methods are and why those methods should be able to address the problem.

Section 3. Proposed High-level Solution and Process (At Least 3 Pages) (10 Points)

In this section, you need to describe and explain your proposed high-level solution and process to solve the problem using the SOTA methods in detail. You may include, but are not limited to:

- A workflow/pipeline diagram to show the entire solution and process, using the SOTA methods, and describe/explain them in detail.
- Your development and implementation for your workflow/pipeline with the SOTA methods. That is, what libraries, modules, classes, functions, etc., have been implemented and used to address your solution and process described above based on your workflow/pipeline.
- Please include any mathematical models, pseudocode algorithms, python code implementations, etc., to support your workflow/pipeline development and implementation.

Section 4. Final Experimental Results and Discussions (At Least 1 Page) (5 Points)

- In this section, you need to describe how you will measure the performance or success of your proposed high-level solution and process using the SOTA methods in detail.
- What data, experiments, metrics, demo, and/or prototypes will you use and conduct, respectively, to assure that your approach(es) are really performing and successful? Please describe them in detail.
- Your final experimental results should include **ALL** SOTA methods for the performance comparisons.
- You can use tables, charts, pictures, etc., to show your final experimental results and provide the discussions.

Section 5. Lessons Learned (At Least 1 Paragraph) (2 Points)

- Describe your experience and what you have learned so far after this project?
- What skills and knowledge you have been practicing or new tools and techniques you are working with, that you did not learn and know before?

Section 6. Conclusions and Future Work (At Least 0.5 Page) (2 Points)

- Describe what your group has done at the end of this project, and whether you have achieved what you set out to do according to the ***task table*** and ***timeline schedule*** in your Phase 1 and Phase 2. That is, list out all the functions/capabilities in the ***task table*** that have been completed.
- List what you have **NOT** managed to accomplish (***if any***) that your group was planning to finish by the end of the semester.
- Give potential tasks for future AI work or to make your product into a really usable and practical system.

References (2 Points)

- Provide a list of expected background material (**AT LEAST 10 Papers from the Google Scholar**: **https://scholar.google.com/**) and/or State-of-the-Art Page at Paper with Code: **https://paperswithcode.com/sota**) that you have planned to read and learn about for your AI project. Those papers should be the recent ones **from 2020 to now**.

Any additional resources, such as a list of manuals, a list of URLs and tutorials, development tools, software environment, system architecture, programming language libraries, and etc., are highly welcome.

- All the references that you list here should have been **cited in Section 1, 2 and other previous sections** already.
- Use the **APA Style** for the references and citations: **https://owl.purdue.edu/owl/research_and_citation/apa_style/apa_style_introduction.html**.

Grading

The grading will primarily be based on the clarity of your final project report in terms of the deliverables described above. Overall, your understanding of the project, its anticipated scope, timeline, etc., should be clearly articulated. A feasibility assessment is important. In some cases, the instructor may call a team meeting to get additional information beyond the submitted final project report, if the report is not clear enough. The readability and professional quality of your document will also be taken into account when assigning the grade.

(A2) Written Final Project Report Content

Option 2 - Innovative/Advanced AI Project

Project Title (1 Point)
Team Members (1 Point)
Photos (1 Point)
Current Programs (1 Point)

Abstract (200 words) (2 Points)

It summarizes the major aspects of the entire AI project in a prescribed sequence that includes:

- The overall purpose of the AI project that your group is working on is to solve a problem.
- The approach to address the problem in your project.
- The current major results and milestones of your project

Keywords (5-10 words or phrases) (1 Point)

Words or phrases in concepts are important in your project.

Section 1. Introduction (At Least 1 Page) (3 Points)

1 Paragraph

- What are the motivational background and context of your domain problem addressed by your project?
- Why is this domain problem important to solve? Please provide some real-world examples and significant statistics with citations to support the importance of this domain problem.

At LEAST 2 Paragraphs

- Describe and explain **at least two** (if your team has two members) or **at least three** (if your team has three members) current state-of-the-art (SOTA) methods with citations that have already addressed this domain problem.
- In your description, please explain what **advantages** of those methods are to address the problem and what **disadvantages** of those methods are and what **gaps** are missing.
- That is, which part(s) of the problem still have **NOT** been considered and solved by using those SOTA methods?

Note: Those existing SOTA methods should be the methods that have been applied to solve the problems described in the most recent papers, i.e., **from 2020 to present**.

At LEAST 2 Paragraphs

- What is/are your advanced/novel approach(es) that you are now proposing to address the above disadvantages and gaps of those current SOTA methods listed above that have not been able to completely solve the problem yet?
- Specifically, please provide a summary contribution of your AI work tasks to address those disadvantages and gaps.

- ✓ What is your proposed high-level solution and process to solve the problem? What differences could you make by using your advanced/novel approach(es) in the problem that you are solving now?
- ✓ How will you evaluate the effectiveness of your approach(es)? That is, you need to describe how you will measure the performance or success of your approach(es) and be able to **replicate** those SOTA approaches described before so that you can compare your own method with those SOTA's in terms of the same performance metrics?
- ✓ What data, experiments, metrics, demo, and/or prototypes will you use to assure your approach(es) really successful that outperforms those SOTA approaches?

Note: **"Novel/Advanced methodologies"** means that they cannot be found in **any existing SOTA approaches** or **any advancement/enhancement** that you can make on top of **any existing SOTA approaches** after doing the literature review. Fine-turning model parameters, training more datasets, applying current SOTA methods on a different domain dataset, etc., by using the same, existing approaches are not counted towards in this category.

1 Paragraph: Outline the subsequent sections that you will include for the rest of your report.

Section 2. SOTA Literature Review (At Least 0.5 Page) (4 Points)

Use **at least one or two paragraphs** to deeply describe each SOTA method that you mention in Section 1. For example, if there are three SOTA methods listed in Section 1, you should have at least three paragraphs, i.e., one paragraph with a workflow diagram for each method. Please also re-state what **disadvantages** of those methods are and what **gaps** are missing. That is, which part(s) of the problems still have **NOT** been considered and solved by using those SOTA methods?

Section 3. Proposed Methodology (At Least 3 Pages) (10 Points)

In this section, you need to describe and explain your proposed approach(es) in detail that you have discussed in Section 1. That may include the architecture, workflow, mathematical formulations for learning models/algorithms, code implementation/library, etc., whatever that you need to develop and implement your approach(es) to bridge the above disadvantages and challenging gaps of the problems that you have stated in Section 2.

Section 4. Final Experimental Results and Discussions (At Least 1 Page) (5 Points)

- In this section, you need to describe how you will measure the performance or success of your approach(es) in detail.
- What data, experiments, metrics, demo, and/or prototypes will you use and conduct, respectively, to assure that your approach(es) are really performing and successful? Please describe them in detail.
- Your final experimental results should include both **ALL** SOTA methods in Section 2 and **your proposed methods** in Section 3 for the performance comparisons.
- You can use tables, charts, pictures, etc., to show your final experimental results and provide the discussions.

Section 5. Lessons Learned (At Least 1 Paragraph) (2 Points)

- Describe your experience and what you have learned so far after this project?
- What skills and knowledge you have been practicing or new tools and techniques you are working with, that you did not learn and know before?

Section 6. Conclusions and Future Work (At Least 0.5 Page) (2 Points)

- Describe what your group has done at the end of this project, and whether you have achieved what you set out to do according to the ***task table*** and ***timeline schedule*** in your Phase 1 and Phase 2. That is, list out all the functions/capabilities in the ***task table*** that have been completed.
- List what you have **NOT** managed to accomplish (***if any***) that your group was planning to finish by the end of the semester.
- Give potential tasks for future AI work or to make your product into a really usable and practical system.

References (2 Points)

- Provide a list of expected background material (**AT LEAST 10 Papers from the Google Scholar**: https://scholar.google.com/) and/or State-of-the-Art Page at Paper with Code: https://paperswithcode.com/sota) that you have planned to read and learn about for your AI project. Those papers should be the recent ones **from 2020 to now**. Any additional resources, such as a list of manuals, a list of URLs and tutorials, development tools, software environment, system architecture, programming language libraries, and etc., are highly welcome.

- All the references that you list here should have been **cited in Section 1, 2 and other previous sections** already.
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Grading

The grading will primarily be based on the clarity of your project progress report in terms of the deliverables described above. Overall, your understanding of the project, its anticipated scope, timeline, etc., should be clearly articulated. A preliminary feasibility assessment is important. In some cases, the instructor may call a team meeting to get additional information beyond the submitted project progress report, if the report is not clear enough. The readability and professional quality of your document will also be taken into account when assigning the grade.

10% bonus points (extra credits) will be added to **your final project score** by the end of the semester if your team will choose this project category, **Option 2**, and also show that your proposed methodology will outperform the existing SOTA approaches to solve the real-world problem in terms of accuracy, speed, and/or other related performance metrics.